Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Lists and Dataclasses will definitely help with keeping track of each person, and their list of gifts. I want and need to improve on coding drafter to actually be able to make the website.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of the project is going to be appending the gift to the specific person's gift list because we have done similar things in other projects / practice problems before, and it's just a use of nested dataclasses and lists of dataclasses. The hardest part is going to be seeing if the budget goes over and organizing the gifts in order.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The Route and States are going to take the longest and the most effort because its something that I am going to have to learn on my own and really practice with. I think that I am also going to get help from an LLM at this part. I want to code the dataclasses first so everything can be worked into the specific person a gift is bought for.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

The distractions that are going to interfere with my work are my other classes assignments that are also due every Friday. But I think I am going to try and start the coding assignment for the project very early every week so I don't have to cram it all in Friday night.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

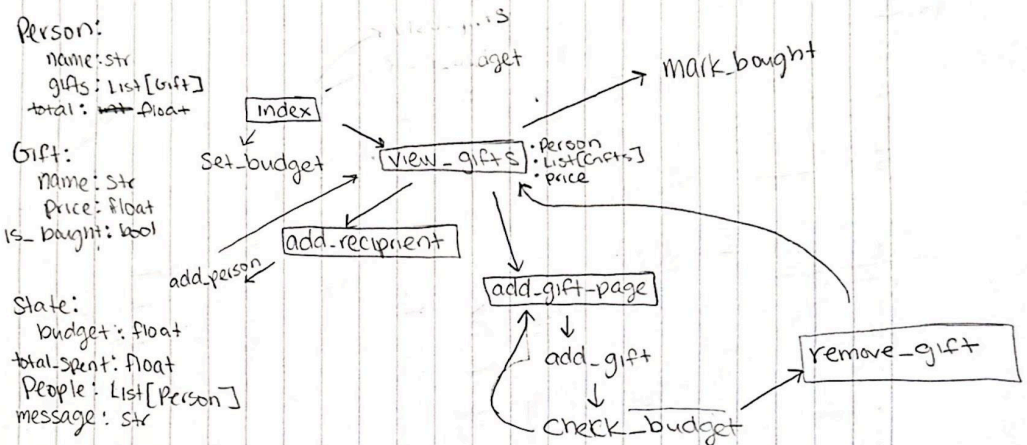
In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Person:
name: str
gifts: List[Gift]
total: ~~str~~ float

Gift:
name: str
price: float
is_bought: bool

State:
budget: float
total_spent: float
People: List[Person]
message: str



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**My website is a Christmas gift planner that tracks the gifts someone wants to buy for everyone in their life. It tracks the total cost and total money spent.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	budget(set once) Total_spent (going to be changed as more gets spent) Gifts (list[Gift]) Gifts_bought (checklist of what was actually bought)
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Gift: Name:str (will be the name of the gift) Price: float Is_bought:bool Person: Name:str (name of the recipient) Gifts: list[Gift] (is a mutable list of every possible gift idea for the person) Total:float (the total price of all the gifts just for this specific person)

Routes (minimum 7) The names of your route functions	Index, set_budget, view_gifts, mark_bought, add_recipreient, add_person, add_gift_page, add_gift, check_budget, remove_gift
Pages (minimum 5) A brief description of the pages in your site	Index (which is the starting page that will prompt user to set a budget) View_gifts (which will show a list of gifts for each person - the home page) Add_reciprient (which will allow the user to add someone they are buying gifts for) Add_gift_page (which will let the user add a gift to a persons gift list) Remove_gift (which will ask the suer what gift they want to remove from the list if they are over budget)
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	The user will have to put in the budget from a textbox, they will have to check if the gift was bought with a checkBox, and add the name of the gift, its price, and who its for with both TextBoxes and SelectBoxes.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	- A check to see if the user has gone over budget because if they have, they need to go to the remove_gift page - A check to see if the gift was bought or not to actually check the item off the checklist - A check to see if any of the lists are empty to total up the overall total for all gifts for all people.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in	- A for loop to iterate over the list[Gift] to total up the prices is going to be needed.

your State attribute. Description of this loop and why it is meaningful to the site	

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM for this assignment since planning is very important to do myself.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.